

Oceans and seas

Oceans are the vast expanses of water filling the areas of low-lying oceanic crust that separate the continents. The shallow edges of the oceans form coastal seas. About 97 percent of the world's water is contained in the oceans. It has been made salty over time by minerals washed off the land by rivers.

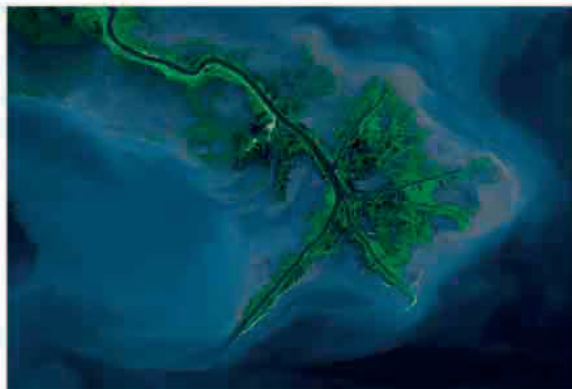
WATER

THE GLOBAL OCEAN

The earliest ocean probably covered the whole globe. Most of its water was originally contained in the rock that formed the planet. Heat turned this water into gas, which erupted as vapor from volcanoes more than 4 billion years ago. As the planet cooled, the vapor turned to rain, which poured down for millions of years, filling the ocean.

▼ DEEP STEAM

Volcanoes, such as this one on the island of Java in Indonesia, still erupt vast amounts of water vapor from deep within the planet.



SALTWATER

After the first continents were formed, rain pouring down on the land weathered the rocks and carried dissolved minerals into the seas. The minerals included sodium chloride, which makes seawater taste salty. This process is still happening today, although the level of salt in the oceans is stable because the extra salt is locked up in ocean-floor rocks.

◀ MINERAL FLOW

This satellite view of the Mississippi Delta shows brown river sediment rich in minerals pouring into the Gulf of Mexico.